

MCA Department - Master of Computer Applications

Course: Advanced Software Engineering (Unit 4: Quality & Process Models)

Topic: Capability Maturity Model Integration (CMMI)

Overview:

CMMI is a process level improvement training and appraisal program developed by Carnegie Mellon University.

It provides organizations with the essential elements of effective processes that ultimately improve their performance.

CMMI models provide guidance for developing or improving processes that meet the business goals of an organization.

CMMI Maturity Levels:

Level 1 - Initial:

Processes are unpredictable, poorly controlled, and reactive. Work gets completed, but it frequently exceeds budget and schedule.

Level 2 - Managed:

Processes are planned, performed, measured, and controlled at the project level. Requirements are managed and controlled.

Level 3 - Defined:

Processes are well characterized and understood, and are described in standards, procedures, tools, and methods.

Level 4 - Quantitatively Managed:

The organization and projects establish quantitative objectives for quality and process performance and use statistical methods to monitor and control the processes.

Level 5 - Optimizing:

Continual process improvement based on a quantitative understanding of business objectives and performance.

Differences between CMMI Staged vs Continuous Representations:

1. Staged Representation:

- Uses predefined sets of process areas to define an improvement path characterized by 5 maturity levels.
- Provides a proven sequence of improvements, each serving as a foundation for the next.

2. Continuous Representation:

- Enables an organization to select a specific process area and improve processes related to it.
- Uses 4 capability levels (Incomplete, Performed, Managed, Defined) to measure individual process improvement.